BANK CUSTOMER CHURN ANALYSIS USING PYTHON

An Exploratory and Statistical Investigation of Customer Attrition Patterns, Banking Behaviour, and Retention Opportunities

Project Documentation

Tools: Python | Pandas | NumPy | Matplotlib | Seaborn | Google Colab

1. Project Overview

This project analyzes bank customer data to explore customer churn patterns and identify characteristics associated with customers leaving the bank. Using Python, the analysis examines customer demographics, geographical differences, account balances, and membership activity to understand how churn varies across customer groups.

The project applies data inspection, data cleaning and transformation, statistical analysis, and data visualization to convert raw customer records into meaningful findings and business recommendations. It is an exploratory and statistical investigation; machine learning and predictive modelling are not included in the completed analysis.

2. Project Objective

To analyze bank customer churn using Python by exploring customer demographics, credit scores, account balances, estimated salaries, and banking behaviour. The project aims to identify customer attrition patterns, examine characteristics associated with churn, and generate evidence-based insights that can support customer-retention decisions.

3. Problem Statement

Customer churn is an important concern in banking because losing customers can affect long-term customer relationships and business performance. This project investigates the question: Which customer characteristics and banking behaviours are associated with higher customer churn, and what retention opportunities can be identified from the available data?

The available dataset records whether customers stayed or exited, but it does not record their stated reasons for leaving. Therefore, the analysis identifies associations and areas for further investigation rather than proving causes of churn.

4. Dataset Information

- Dataset name: Bank Customer Churn
- Source: Maven Analytics Data Playground
- Domain: Banking and Customer Analytics
- File format: CSV
- Dataset size: 10,000 customer records and 13 original columns
- Analysis environment: Google Colab
- Programming language: Python
- Target variable: Exited (0 = Stayed; 1 = Exited)

The dataset contains customer demographic, financial, and banking-behaviour information. The source material does not specify the original customer-data collection period.

5. Dataset Attributes

Attribute	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer credit score
Geography	Customer's country
Gender	Customer gender
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
NumOfProducts	Number of banking products held
HasCrCard	Credit card ownership indicator
IsActiveMember	Active membership indicator
EstimatedSalary	Estimated customer salary
Exited	Customer churn indicator

6. Tools and Technologies

- Python – main programming language
- Pandas – data loading, cleaning, grouping and summaries
- NumPy – numerical operations
- Matplotlib – chart creation
- Seaborn – statistical visualizations
- Google Colab – notebook environment
- GitHub – source-code storage and version control

7. Project Workflow

Stage 1: Initial Exploratory Data Analysis

The dataset was loaded from GitHub into a Pandas DataFrame. Initial exploration included previewing records, checking dataset dimensions, reviewing column names and data types, generating descriptive statistics, and checking missing values, duplicate records, and the target-variable distribution.

Stage 2: Data Cleaning, Transformation and Feature Engineering

The dataset was checked for duplicate records and missing values. No duplicate rows or missing values were found in the original dataset. Column names were standardized to lowercase with underscores. The data types were inspected, and two derived features were created: churn_status (Stayed/Exited) and age_group. The final dataset contained 10,000 rows and 15 columns, with no missing values or duplicate rows.

Stage 3: Statistical Analysis and Visualization

Statistical analysis included measures of central tendency, variance, standard deviation, skewness, kurtosis, and age outlier detection using a box plot and the IQR method. Nine visualizations were used to explore churn distribution, age, geography, membership activity, account balances, and combinations of age group, geography, and membership activity.

Stage 4: Documentation, Insights and Presentation

The findings are summarized in plain English, with business implications and recommendations. The analysis distinguishes observed associations from confirmed causes and identifies predictive modelling as a possible future extension.

8. Data Preparation and Statistical Analysis

8.1 Central Tendency

Variable	Mean	Median	Mode
Credit Score	650.53	652.00	850
Age	38.92	37.00	37
Account Balance	76,485.89	97,198.54	0.00
Estimated Salary	100,090.24	100,193.91	24,924.92

Interpretation: The average customer age is approximately 39 years. The mean and median credit scores and estimated salaries are close. The mean account balance is below its median, while the mode is zero, indicating that zero balances occur frequently. These statistics describe the overall customer base; they do not by themselves explain churn.

8.2 Variance and Standard Deviation

Variable	Variance	Standard Deviation
Credit Score	9,341.86	96.65
Age	109.99	10.49
Account Balance	3,893,436,175.99	62,397.41
Estimated Salary	3,307,456,784.13	57,510.49

Interpretation: Customer ages typically vary by about 10.49 years around the mean age. Credit scores have a standard deviation of 96.65 points. Account balances and estimated salaries also show substantial spread in their original units. Standard deviations should not be compared directly across variables with different units and scales.

8.3 Skewness, Kurtosis and Outlier Detection

Variable	Skewness	Excess Kurtosis
Credit Score	-0.072	-0.426
Age	1.011	1.395
Account Balance	-0.141	-1.489
Estimated Salary	0.002	-1.182

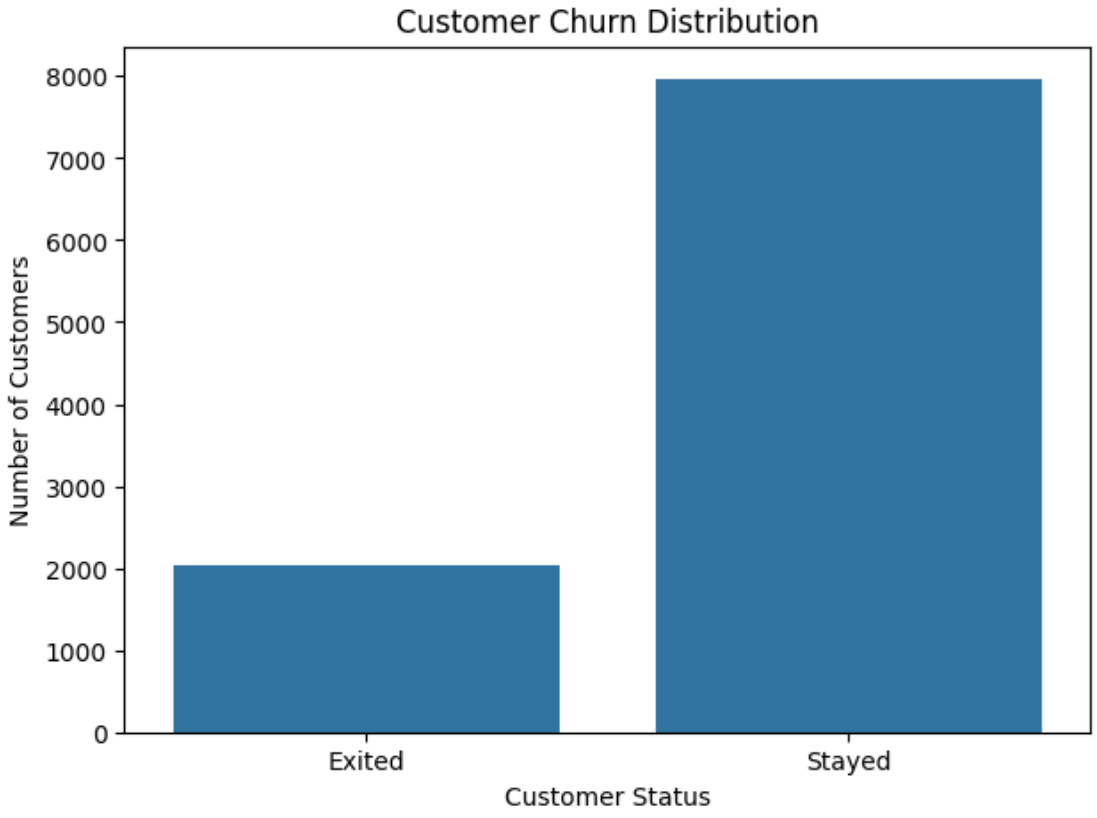
Interpretation: Age has positive skewness (1.011), indicating a longer right tail toward older customers. Credit score and account balance are close to symmetric, with slight negative skew. Estimated salary is almost symmetric. The negative excess kurtosis values for credit score, balance and salary indicate flatter distributions with lighter tails than a normal distribution; age's positive excess kurtosis indicates heavier tails and a more pronounced peak.

The age IQR analysis produced Q1 = 32, Q3 = 44, IQR = 12, a lower bound of 14 and an upper bound of 62. It identified 359 potential age outliers above 62. These records were retained because they may represent valid customers; statistical outliers are not automatically errors.

9. Visualizations and Interpretations

The following figures are screenshots of the charts generated in the submitted Google Colab notebook. Each figure is followed by its interpretation and business relevance.

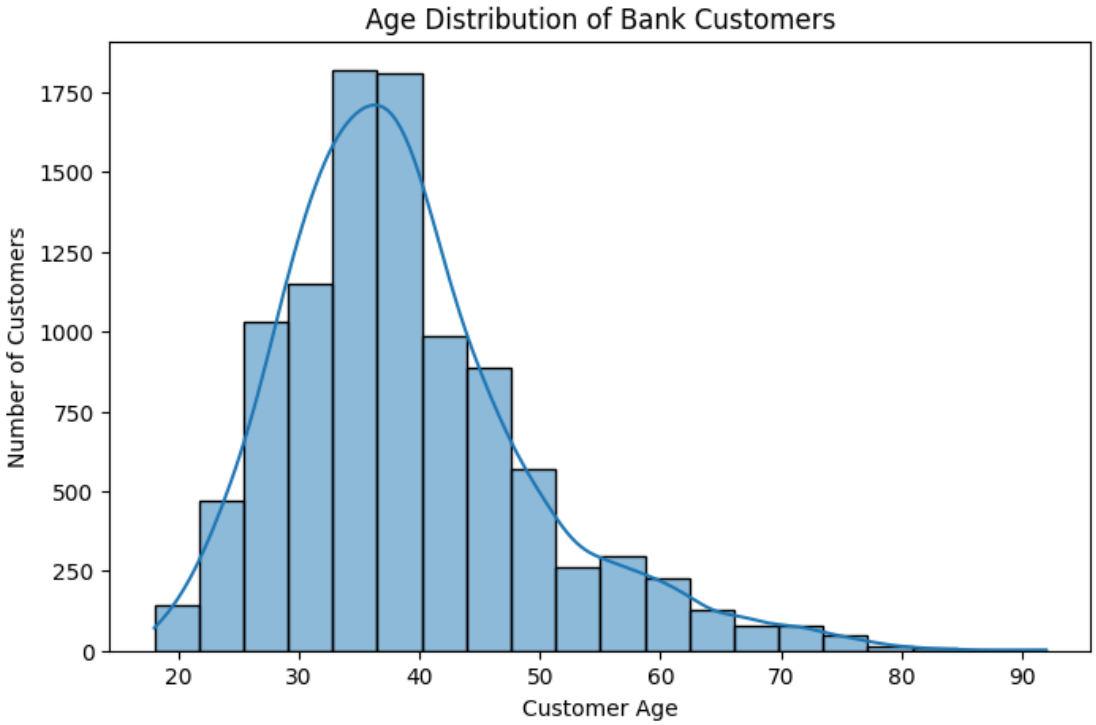
Figure 1. Customer Churn Distribution



Interpretation: The count plot shows 7,963 customers (79.63%) stayed and 2,037 (20.37%) exited. The overall churn rate is 20.37%, or approximately one in five customers in this dataset.

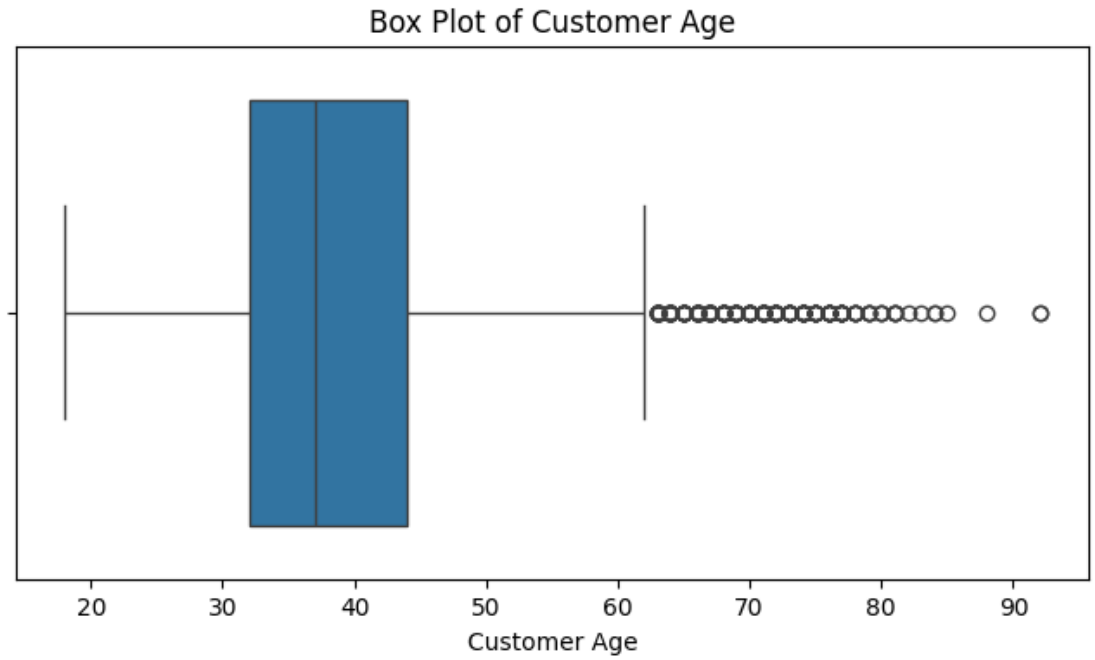
Business relevance: This overall churn rate is the baseline against which segment-level churn rates can be compared.

Figure 2. Age Distribution of Customers



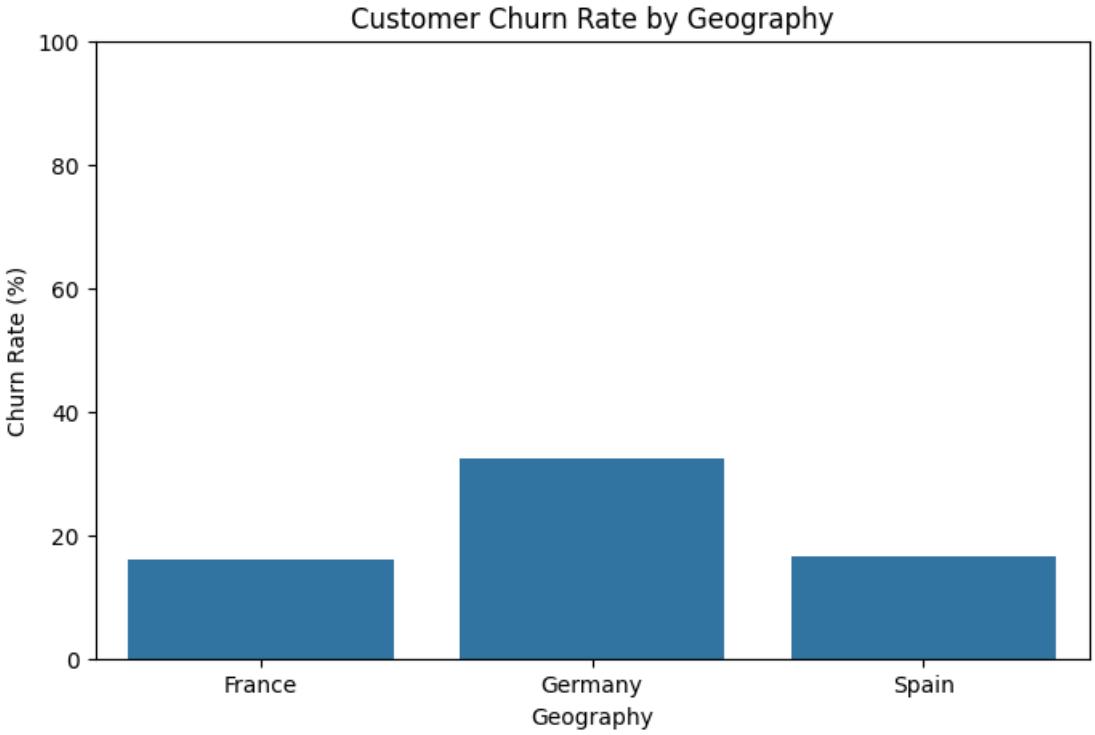
Interpretation: The histogram shows the distribution of customer ages. The typical customer is in the late 30s (mean 38.92 years; median 37 years). The age distribution is positively skewed, indicating a longer tail toward older customers.

Figure 3. Box Plot of Customer Age



Interpretation: The box plot summarizes the median and spread of customer ages and highlights potential high-age outliers. The IQR analysis found Q1 = 32, Q3 = 44, IQR = 12, lower bound = 14 and upper bound = 62. It identified 359 potential age outliers above 62. These were retained because they may be valid customers.

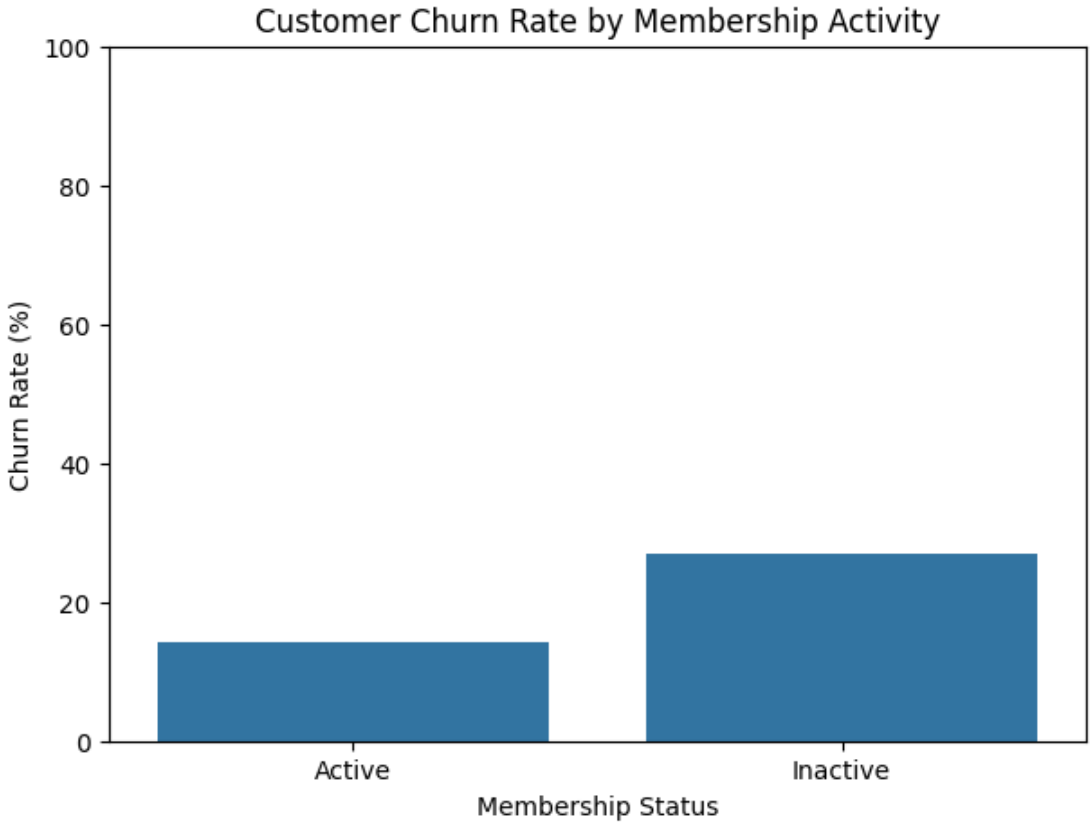
Figure 4. Customer Churn Rate by Geography



Interpretation: Germany had the highest overall churn rate at 32.44%, compared with Spain (16.67%) and France (16.15%). This identifies Germany as a priority for further investigation; the chart does not explain the reasons for the difference.

Business relevance: The geographical difference suggests that retention conditions or customer experiences may vary by country and should be investigated.

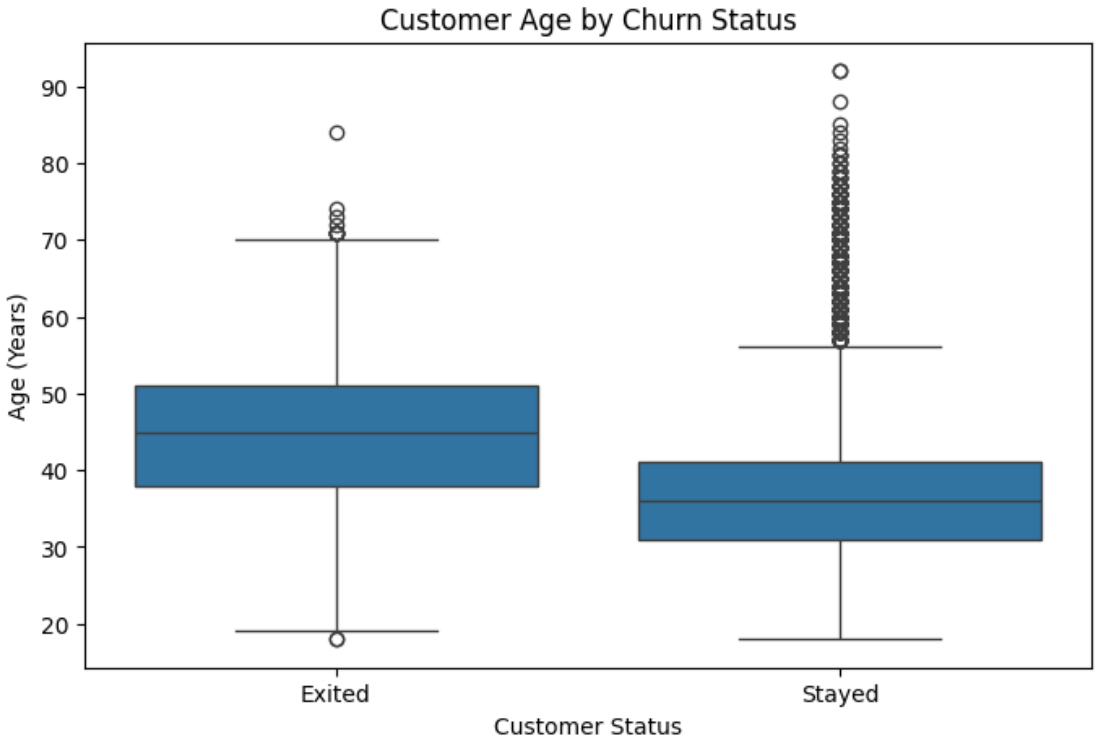
Figure 5. Customer Churn Rate by Active Membership



Interpretation: Inactive members had a churn rate of 26.85%, compared with 14.27% among active members, a difference of 12.58 percentage points. This indicates an association between inactivity and higher churn.

Business relevance: The association supports examining engagement levels and testing ways to improve customer engagement.

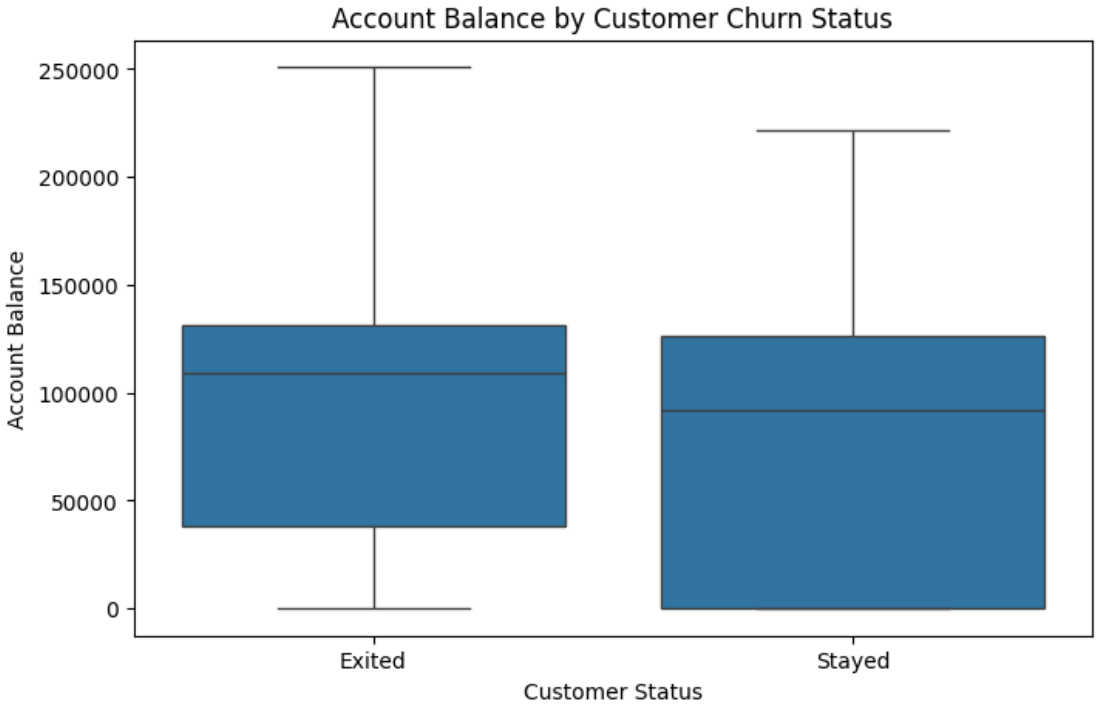
Figure 6. Customer Age by Churn Status



Interpretation: Exited customers were older on average (44.84 years; median 45) than customers who stayed (37.41 years; median 36). This shows an age difference between the two groups, but does not establish age as a cause of churn.

Business relevance: The age difference suggests that customer needs and experiences may vary by age segment.

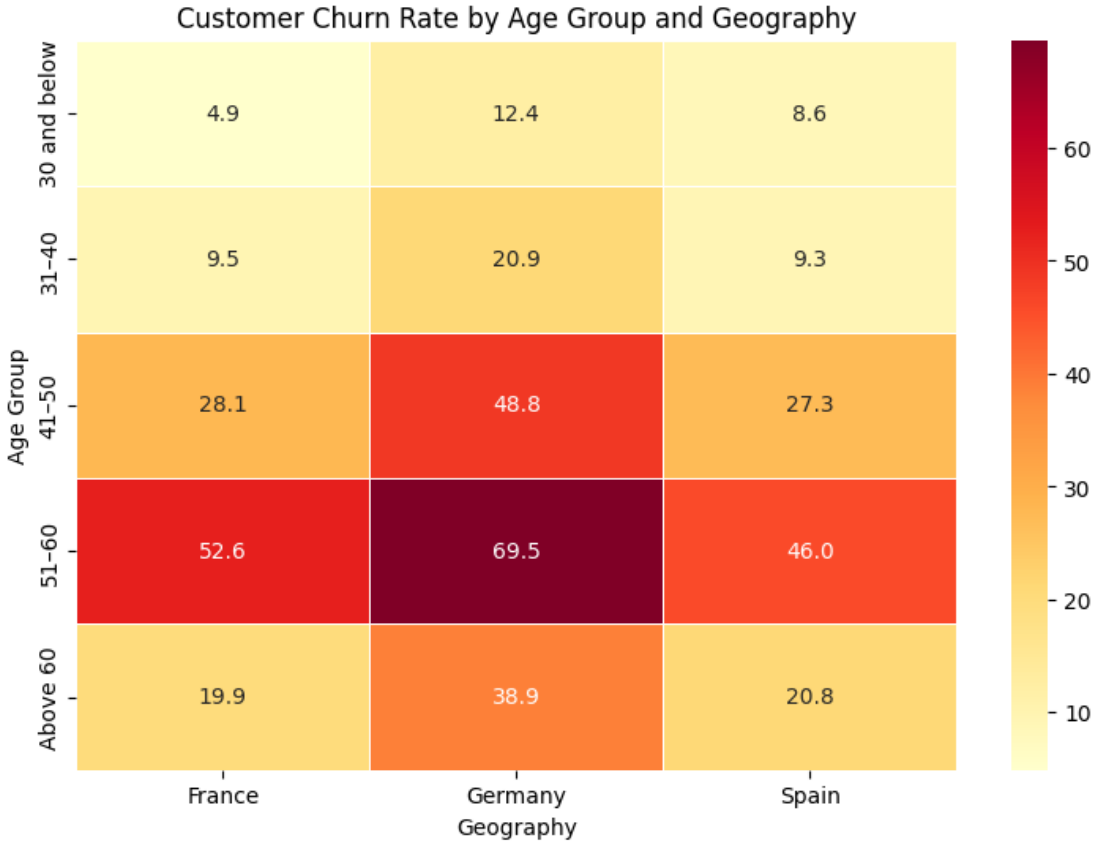
Figure 7. Account Balance by Churn Status



Interpretation: Exited customers had a higher average balance (91,108.54) and median balance (109,349.29) than customers who stayed (average 72,745.30; median 92,072.68). This pattern warrants further investigation and does not mean higher balances cause churn.

Business relevance: Higher balances among exited customers make this segment worth examining, but the chart does not reveal the reason for exit.

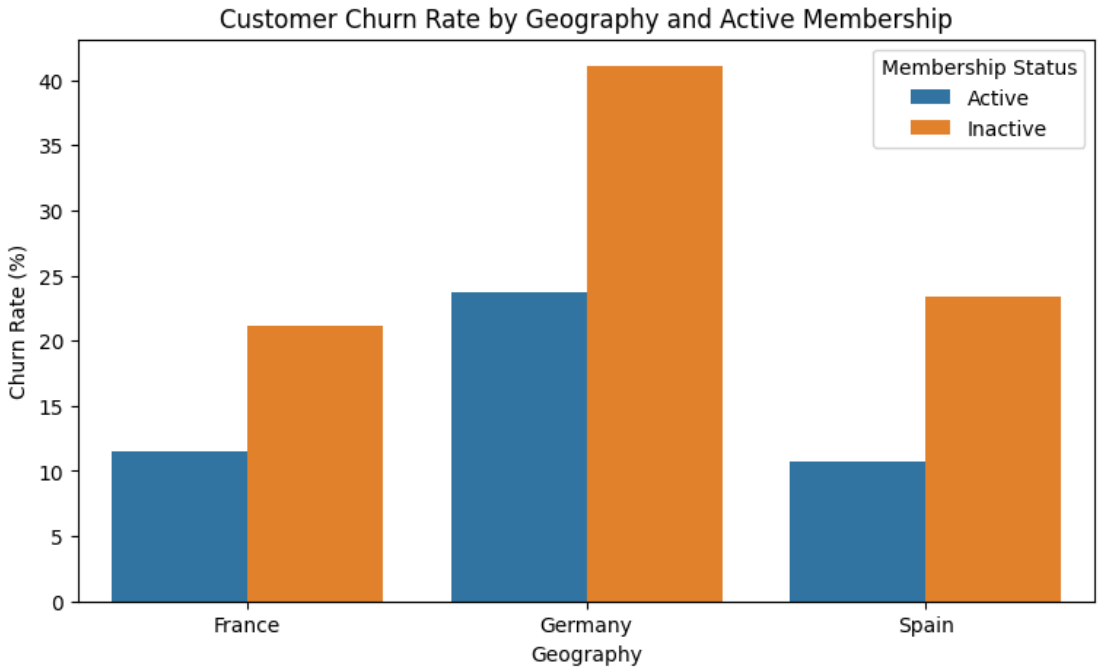
Figure 8. Churn Rate by Age Group and Geography



Interpretation: The highest observed age-geography churn rate was among customers aged 51–60 in Germany (69.55%). Other elevated rates were 52.59% for ages 51–60 in France and 48.82% for ages 41–50 in Germany. Group sizes should be checked before prioritizing segments.

Business relevance: This segment is a candidate for focused investigation. Segment customer counts should be checked because a high percentage from a small group may be less stable.

Figure 9. Churn Rate by Geography and Active Membership



Interpretation: Inactive members had higher churn rates in all three countries. Germany recorded 41.08% for inactive and 23.72% for active members; Spain recorded 23.35% and 10.75%; France recorded 21.13% and 11.50%, respectively.

Business relevance: The pattern is consistent across all three countries, making membership activity a useful dimension for further retention analysis.

10. Key Findings

- Overall churn: 2,037 of 10,000 customers exited (20.37%); 7,963 stayed (79.63%).
- Geography: Germany had the highest overall churn rate (32.44%), compared with Spain (16.67%) and France (16.15%).
- Membership activity: Inactive members had a 26.85% churn rate, compared with 14.27% among active members, a difference of 12.58 percentage points.
- Age: Exited customers had an average age of 44.84 years and median age of 45, compared with 37.41 years and median age of 36 among customers who stayed.
- Account balance: Exited customers had a higher average balance (91,108.54) and median (109,349.29) than customers who stayed (average 72,745.30; median 92,072.68).
- Age and geography: Customers aged 51–60 in Germany had the highest observed age-geography churn rate (69.55%). Other elevated rates included ages 51–60 in France (52.59%) and ages 41–50 in Germany (48.82%).
- Geography and activity: Inactive members had higher churn in all countries. Germany recorded 41.08% for inactive members and 23.72% for active members; Spain 23.35% and 10.75%; France 21.13% and 11.50%.

11. Business Recommendations

1. Investigate retention challenges in Germany, where the overall churn rate is highest. Review customer experiences, product suitability, service expectations and local market conditions.

2. Investigate why inactive members have higher churn rates. Consider testing relevant engagement initiatives, personalized communication and product recommendations.
3. Examine the needs and experiences of customers aged 41–60, especially in Germany, while checking the number of customers in each segment.
4. Investigate higher-balance customers who exited to understand whether their service needs or banking experiences differ.
5. Monitor overall and segment-level churn rates over time and evaluate retention initiatives using subsequent customer data.

12. Limitations

- The dataset does not state the customers' reasons for leaving; therefore, the analysis cannot identify exact causes of churn.
- The analysis identifies associations and group differences, not causal relationships.
- The original data collection period is not specified in the supplied dataset documentation.
- Some age-geography combinations may contain fewer customers than others; group sizes should be reviewed before prioritizing action.
- The project does not include a predictive machine learning model or a model evaluation.

13. Future Enhancements

- Add customer transactions, complaints, service interactions, product usage and exit-survey data to investigate reasons for churn more directly.
- Use a larger or more recent dataset and monitor churn over time.
- Conduct statistical significance tests for selected group comparisons.
- Build an interactive dashboard in Power BI or Tableau.
- Develop and evaluate classification models such as Logistic Regression, Decision Tree and Random Forest using appropriate metrics including precision, recall, F1-score, ROC-AUC and a confusion matrix.
- Test customer-retention initiatives and measure their outcomes.

14. Conclusion

The Bank Customer Churn Analysis project demonstrates how Python, statistical analysis and visualization can be used to investigate customer attrition patterns. The dataset contains 10,000 customer records and has an overall churn rate of 20.37%. The analysis identifies higher churn in Germany (32.44%), among inactive members (26.85%), and among selected age groups. Customers who exited were generally older and had higher average and median account balances than customers who stayed.

The combined age-group and geography analysis highlighted customers aged 51–60 in Germany, with an observed churn rate of 69.55%, as a segment warranting further investigation. Inactive members in Germany also showed a high churn rate of 41.08%. These patterns can help the bank decide where to investigate customer experiences and retention opportunities.

The dataset does not record why customers left, so the findings should be interpreted as associations rather than proven causes. Overall, the project demonstrates practical skills in data inspection, cleaning, transformation, exploratory data analysis, statistical interpretation, outlier detection, visualization and business-focused reporting. Future work could incorporate customer feedback and transaction data, test retention interventions, and develop predictive churn models.